COMPREHENSIVE DATA SCIENCE ANALYSIS OF LAPD CRIME DATA (2020–2025)

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Date: May 4, 2026

Topic: Precision Medicine and Healthcare Analytics Portfolio (LAPD Dataset Case Study)

1. Introduction

Urban crime analysis serves as a vital pillar of modern municipal management, influencing the distribution of patrol units and the long-term drafting of social policy. The City of Los Angeles, one of the largest and most diverse metropolitan areas in the United States, presents a complex landscape for public safety. This report leverages the Los Angeles Police Department (LAPD) Open Data Portal to analyze over one million reported crime incidents spanning January 2020 to May 2025 [cite: 220, 237].

Business Problem Statement

The core challenge facing law enforcement is the efficient allocation of finite resources to combat a high volume of incidents. Traditional reactive policing often falls short in identifying evolving trends or predicting high-severity events. The objective of this analysis is to transform raw incident reports into actionable intelligence by identifying spatial and temporal hotspots, predicting crime severity with high accuracy, and utilizing natural language processing (NLP) to categorize unstructured narratives. By doing so, the project aims to support evidence-based decision-making and enhance proactive public safety interventions [cite: 227, 228].

2. Technical Analysis

2.1 Descriptive Analysis and Preprocessing

The analysis began with a dataset of 1,004,991 records [cite: 237]. Preprocessing involved engineering temporal features such as Time of Day (Morning, Afternoon, Evening, Night) and cleaning victim demographics, including age (filtered for ranges 1–100) and mapped sex/descent labels [cite: 240, 241, 242]. Crimes were categorized into broader types like "Property Crime" and "Violent Crime" and mapped to severity levels based on FBI classifications (Part 1 and Part 2) [cite: 243, 244].

2.2 Exploratory Data Analysis (EDA)

The EDA revealed that property crimes overwhelmingly dominate the landscape at 57.9%, followed by violent crimes at 20.9% [cite: 249]. Analysis showed that crimes peak during evening hours (17:00–21:00) and are highest on Fridays and Saturdays [cite: 254, 255]. A longitudinal view highlighted a sharp decline in reported crimes in early 2020 due to COVID-19 lockdowns, followed by a steady climb and stabilization by 2022 [cite: 256, 257]. Geographically, areas such as "Central" and "77th Street" reported the highest volumes [cite: 261].

2.3 Machine Learning

We implemented a binary classification task to predict crime severity. XGBoost emerged as the superior model, achieving 85.3% accuracy [cite: 268, 269].

Model	Accuracy	AUC-ROC	F1-Score
Logistic Regression	0.6645	0.7964	0.7569
Random Forest	0.8398	0.9293	0.8634
XGBoost	0.8530	0.9387	0.8737

K-Means clustering (***k*=5**) further segmented incidents into profiles like "Late Night Central Crimes" (high severity/weapon use) and "Midday Suburban Property Crimes" [cite: 273, 274].

2.4 Time Series Forecasting

Monthly volume was forecasted using Prophet and SARIMA models. Prophet outperformed SARIMA, providing robust projections that captured underlying seasonal peaks in summer months (July–August) [cite: 284, 288].

3. Research and Peer-Reviewed Synthesis

Article 1: Exploring the Immediate Effects of COVID-19 Containment Policies on Crime (Campedelli et al., 2020)

Campedelli et al. utilized Bayesian structural time-series to investigate the immediate impact of COVID-19 containment in Los Angeles. Their research found that overall crime significantly decreased—specifically robbery, shoplifting, and theft—while categories like vehicle theft and assault with deadly weapons saw no significant initial effect. This suggests that "stay-at-home" orders primarily disrupted opportunity-based crimes in public spaces.

Research Application: This research directly supports our longitudinal EDA, which identified the same sharp decline in early 2020 [cite: 256]. By understanding the "opportunity reduction" theory presented by Campedelli, our model's feature importance (which values "Premise" and "Time") can be contextualized as a measurement of changing routine activities post-pandemic.

Article 2: Down with the Sickness? Los Angeles Burglary and COVID-19 (Ashby, 2020)

Ashby's research used seasonal ARIMA to analyze crime trends across 16 American cities, focusing on how lockdowns shifted crime spatiality. The study noted that while residential burglaries decreased as guardians stayed home, commercial areas often saw crime shifts. It emphasizes that crime patterns are not static and are deeply influenced by shifts in socioeconomic "routine activities."

Research Application: Our analysis incorporates similar time-series methodologies (Prophet and SARIMA) [cite: 277]. Ashby's focus on the "guardian" effect explains why our clustering identified distinct profiles for midday suburban vs. central night crimes [cite: 274], allowing us to refine our predictive models to account for these environmental variations.

Article 3: Automated Classification of Crime Narratives (NSI, 2025)

This research evaluated different machine learning strategies, including XGBoost and BERT, for the automatic codification of crime narratives. It demonstrated that language models significantly improve coding consistency for official statistics. The study highlighted that while deep learning (BERT) is accurate, algorithms like XGBoost remain highly effective for structured data classification in production environments.

Research Application: Our project mirrors this by using XGBoost as the primary classification model, achieving 85.3% accuracy [cite: 268]. Additionally, our NLP section [cite: 292] used LDA topic modeling to categorize unstructured descriptions into themes like "Identity Theft" and "Domestic Violence," applying the "semantic buckets" concept suggested by this scholarly work [cite: 317].

4. Interactive Dashboard

To facilitate stakeholder engagement, an interactive Plotly Dash dashboard was developed. This tool serves as a visual bridge between complex data models and actionable policy, providing high-level KPIs such as total incident counts and weapon usage statistics [cite: 306, 307]. Users can dynamically filter data by Year, LAPD Area, and Crime Category, enabling localized drill-downs into specific neighborhoods [cite: 308]. The interface includes Mapbox-powered geographic scatter maps and seasonal heatmaps that allow users to visualize hotspots in real-time [cite: 309]. By transforming over one million records into a responsive environment, the dashboard ensures that predictive insights—such as severity classifications and seasonal forecasts—remain accessible to non-technical decision-makers, thereby supporting evidence-based resource deployment across the city [cite: 310].

5. Conclusion

This comprehensive analysis of the LAPD crime dataset (2020–2025) demonstrates the efficacy of integrating traditional statistical methods with modern machine learning and NLP techniques. By analyzing over one million records, we established that property crimes overwhelmingly dominate the Los Angeles landscape, while violent incidents require distinct, resource-intensive policing strategies [cite: 313, 314]. Our XGBoost model proved highly effective at predicting crime severity, achieving an 85.3% accuracy rate by leveraging contextual features such as time, location, and premises [cite: 315]. Furthermore, time series forecasting revealed a stabilization of crime rates post-pandemic, with predictable peaks during the summer months that offer a reliable baseline for seasonal resource scaling [cite: 316]. The use of NLP topic modeling successfully categorized unstructured crime descriptions into semantic themes, providing a deeper understanding of the "why" behind the numbers [cite: 317]. Ultimately, this project provides a data-driven roadmap for proactive public safety, proving that predictive analytics can move law enforcement from reactive responses to anticipatory planning [cite: 312].

6. Recommendations

- **Seasonal Resource Scaling:** LAPD should increase patrol density during the identified July–August seasonal peaks to address predictable surges in incident volume [cite: 288, 316].
- **Targeted Property Crime Prevention:** Given the 57.9% dominance of property crime, city resources should be directed toward community "guardianship" programs in high-volume areas like the Central and 77th Street districts [cite: 249, 261, 314].
- **Socioeconomic Feature Engineering:** Future model iterations should integrate external datasets, such as unemployment rates and housing values, to enhance the precision of spatial clustering and severity predictions [cite: 318].
- **Real-time Integration:** The developed Plotly Dash dashboard should be integrated into daily strategic briefings to allow for real-time monitoring of emerging hotspots and bias identification [cite: 306, 310].

References

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